

Reviewer Pack — Minimal Order of Geometric Entanglement and Communication Co...

Entry eca17cce5175 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Order of Geometric Entanglement and Communication Complexity Rank Correlation

Entry ID: eca17cce5175 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-04 15:50:36 UTC

1. Conjecture

Field A (mathematical branch): Quantum Information Theory (Geometric Entanglement) **Field B** (complexity object): Communication Complexity (Matrix Rank)

Statement:

The minimal order of geometric entanglement of a quantum system is linearly correlated with the communication complexity rank of its associated classical information processing task, such that $O(\text{geo_entangle}(Q)) = \Theta \text{ comm_complexity_rank}(C)$, where $\text{geo_entangle}(Q)$ represents the minimal order of geometric entanglement for a quantum state Q and $\text{comm_complexity_rank}(C)$ represents the communication complexity rank for the classical information processing task C .

Rationale (proposer's reasoning):

Geometric entanglement captures the non-local correlations in quantum systems, which are closely related to the amount of communication needed to simulate classical tasks. This conjecture suggests that the order of geometric entanglement can serve as a lower bound on the communication complexity rank, providing insights into the structure of classical information processing tasks.

Taxonomy category: `geometric_entanglement_comm_complexity_rank` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): b823153bb884ddc1

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if at least 80% of the computed pairs ($\text{geo_entangle}(Q)$, $\text{comm_complexity_rank}(C)$) have a Euclidean distance to their regression line's mean less than or equal to 3 standard deviations, and the Pearson correlation coefficient is greater than 0.5, across 30 random seeds.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "geometric entanglement" AND "communication complexity rank" - "quantum information theory" AND "matrix rank in communication complexity" - "minimal order of geometric entanglement" related to "communication complexity"

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def gaussian_elimination(A):
        m, n = len(A), len(A[0])
        for i in range(m):
            max_row = i
            for j in range(i+1, m):
                if abs(A[j][i]) > abs(A[max_row][i]):
                    max_row = j
            A[i], A[max_row] = A[max_row], A[i]
            factor = A[i][i]
            for j in range(n):
                A[i][j] /= factor
            for j in range(m):
                if j != i:
                    factor = A[j][i]
                    for k in range(n):
                        A[j][k] -= factor * A[i][k]
        return A

    def matrix_multiplication(A, B):
        m, n, p = len(A), len(B), len(B[0])
        C = [[0]*p for _ in range(m)]
        for i in range(m):
            for j in range(p):
                for k in range(n):
```

```

        C[i][j] += A[i][k] * B[k][j]

    return C

def communication_complexity_rank(C):
    n = len(C)
    rank = 0
    while True:
        pivot_row, pivot_col = None, None
        for i in range(n):
            if all(A[i][j] == 0 for j in range(n)):
                continue
            if pivot_row is None or abs(A[i][pivot_col]) > abs(A[pivot_row][pivot_col]):
                pivot_row = i
        if pivot_row is None:
            break
        rank += 1
        factor = A[pivot_row][pivot_col]
        for j in range(n):
            A[pivot_row][j] /= factor
        for i in range(n):
            if i != pivot_row and abs(A[i][pivot_col]) > 0:
                factor = A[i][pivot_col]
                for j in range(n):
                    A[i][j] -= factor * A[pivot_row][j]

    return rank

def geo_entangle(Q):
    # Placeholder function to simulate geometric entanglement calculation
    # This is a dummy implementation and should be replaced with actual logic
    return random.uniform(0, 1)

n = random.randint(5, 40)
Q = [[random.random() for _ in range(n)] for _ in range(n)]
C = [[random.choice([0, 1]) for _ in range(n)] for _ in range(n)]

geo_entangle_value = geo_entangle(Q)
comm_complexity_rank_value = communication_complexity_rank(C)

return {
    "metric_name": "geo_entangle_comm_complexity_corr",
    "metric_value": geo_entangle_value * comm_complexity_rank_value,
    "instances_tested": 1,
    "n_max": n,
    "conjecture_holds": False,
    "counterexample": "mapping_undefined"
}

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)

```

```

std_dev = math.sqrt(sum((result["metric_value"] - mean_value) ** 2 for result in results) / len(results))
support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

if support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_dev} support_fraction={support_fraction}")
elif any(not result["conjecture_holds"] for result in results):
    first_failing_seed = next((seed for seed, result in zip(seeds, results) if not result["conjecture_holds"]), None)
    print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")
else:
    print("RESULT: INCONCLUSIVE reason=mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_26f1fd64.py", line 96, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_26f1fd64.py", line 81, in run_trial
    comm_complexity_rank_value = communication_complexity_rank(C)
                                ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_26f1fd64.py", line 54, in communication_complexity_rank
    if all(A[i][j] == 0 for j in range(n)):
       ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_26f1fd64.py", line 54, in <genexpr>
    if all(A[i][j] == 0 for j in range(n)):
       ^
NameError: name 'A' is not defined

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from evaluating whether the conjecture is supported or falsified according to the pre-registered criteria. | next: Investigate and fix the crash in the test code to proceed with the evaluation of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14734
2	preregistration	ollama_remote	glm4:latest	0	0	12778
3	novelty	ollama_remote	glm4:latest	0	0	8335
4	novelty	ollama_remote	glm4:latest	0	0	8809
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	46071
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12805
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10594
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11244
9	judge	ollama_remote	glm4:latest	0	0	15834

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 141203 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/eca17cce5175.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/eca17cce5175.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/eca17cce5175.tar.gz` (if generated)